Code

# Spatial Transcriptomics - Seurat Analysis of Human Glioblastoma (GBM) FFPE Tissue

## Setting up environment for Spatial Analysis

## Packages/Dependencies

Hide

```
library(Seurat)
library(SeuratData)
library(ggplot2)
library(patchwork)
library(dplyr)
library(Rfast2)
```

## Loading 10x Visium Dataset

Hide

```
data_dir <- "C:/Users/hugor/OneDrive/Desktop/Bioinformatics/Spatial Transcriptomics/Glioblast
oma Spatial Data"

list.files(data_dir)
```

```
[1] "CytAssist_11mm_FFPE_Human_Glioblastoma_metrics_summary.csv" "filtered_feature_bc_matrix.
h5"
[3] "spatial"
```

Hide

```
install.packages("hdf5r")
```

```
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/hdf5r_1.3.16.zip'
Content type 'application/zip' length 4010902 bytes (3.8 MB)
downloaded 3.8 MB
```

```
package ‘hdf5r’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in
    C:\Users\hugor\AppData\Local\Temp\RtmpW8uXqM\downloaded_packages
```

Hide

```
library(hdf5r)

GBM <- Load10X_Spatial(data.dir = data_dir)

GBM
```

```
An object of class Seurat
18085 features across 10878 samples within 1 assay
Active assay: Spatial (18085 features, 0 variable features)
 1 layer present: counts
 1 spatial field of view present: slice1
```

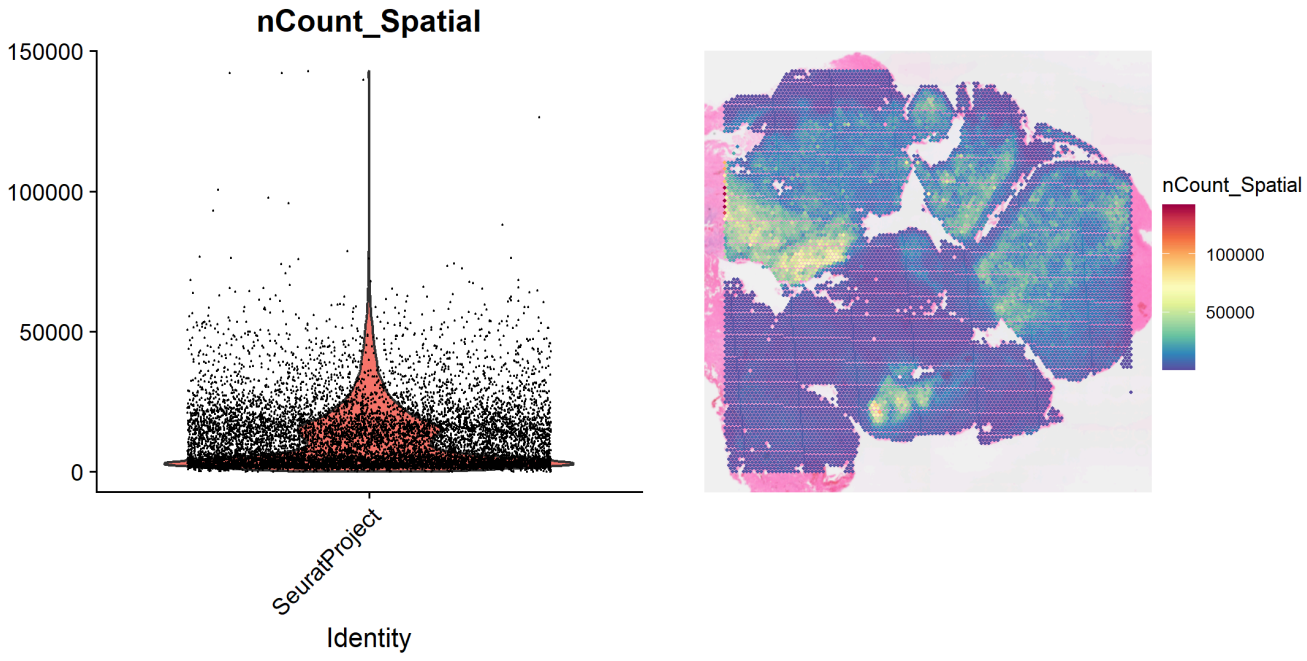
# Data preprocessing

Hide

```
plot1 <- VlnPlot(GBM, features = "nCount_Spatial", pt.size = 0.1) + NoLegend()

plot2 <- SpatialFeaturePlot(GBM, features = "nCount_Spatial", pt.size.factor = 3) + theme(leg
end.position = "right")

wrap_plots(plot1, plot2)
```




# Spatial Normalisation

Hide

```
GBM <- SCTransform(GBM, assay = "Spatial", verbose = FALSE)
```

# Gene expression visualisation

Hide

```
features <- rownames(GBM)

## Sanity check
head(features)
```

```
[1] "SAMD11"  "NOC2L"   "KLHL17"  "PLEKHN1" "PERM1"   "HES4"
```

Hide

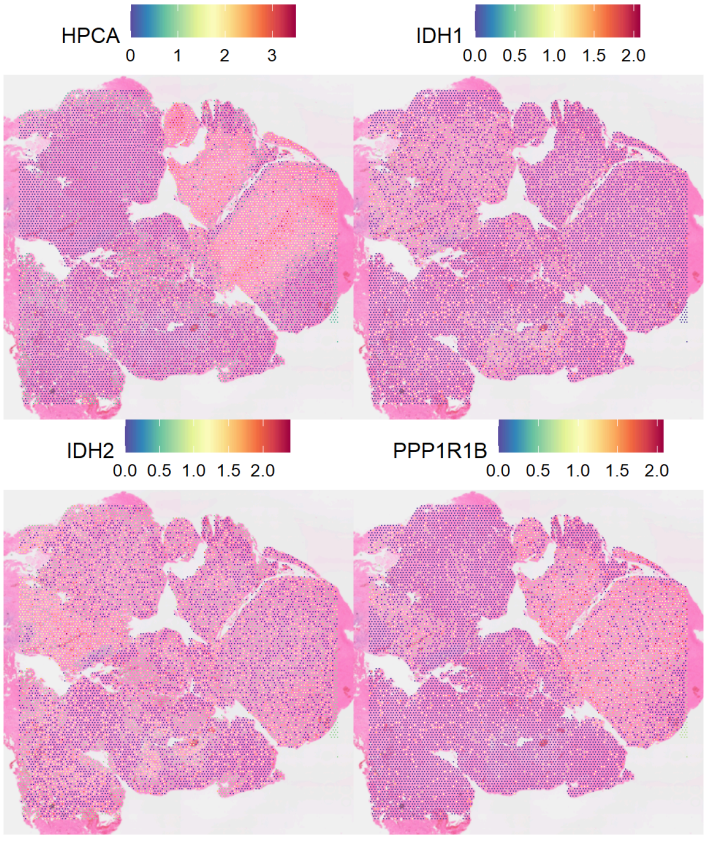
```
'ATG5' %in% features
```

```
[1] TRUE
```

# Visualisation of 4 Genes

Hide

```
SpatialFeaturePlot(GBM, features = c("HPCA", 'IDH1', 'IDH2', 'PPP1R1B'))
```




Hide

```
SpatialFeaturePlot(
  GBM,
  features = c('IDH1'),
  images = NULL,
  crop = TRUE,
  slot = "data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = NULL,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 2,
  image.scale = "lowres",
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
)
```

# Dimensionality reduction, clustering

Hide

```
GBM <- RunPCA(GBM, assay = "SCT", verbose = FALSE)
GBM <- FindNeighbors(GBM, reduction = "pca", dims = 1:30)
GBM <- FindClusters(GBM, verbose = FALSE)
GBM <- RunUMAP(GBM, reduction = "pca", dims = 1:30)
```

```
Using method 'umap'
0%   10   20   30   40   50   60   70   80   90   100%
[----|----|----|----|----|----|----|----|----|----|
**************************************************|
```

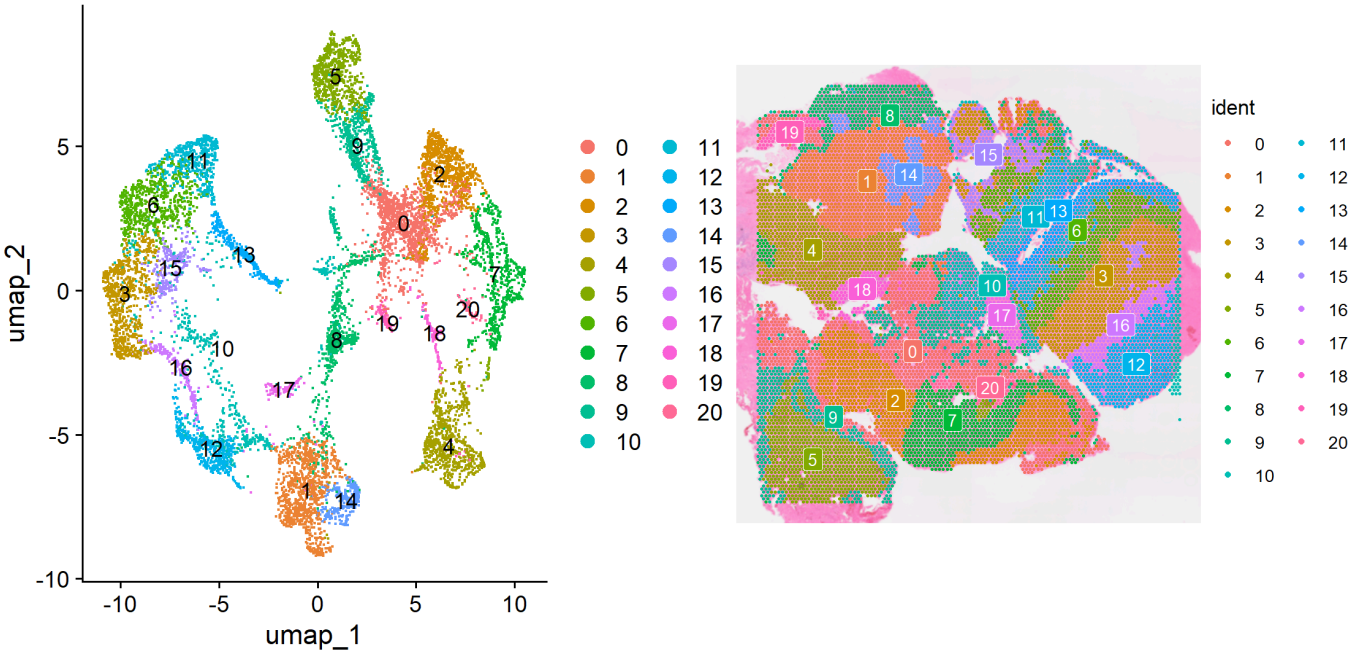
# Visualisation

Hide

```
p1 <- DimPlot(GBM, reduction = "umap", label = TRUE)

p2 <- SpatialDimPlot(
  GBM,
  group.by = NULL,
  images = NULL,
  cols = NULL,
  crop = TRUE,
  cells.highlight = NULL,
  cols.highlight = c("#DE2D26", "grey50"),
  facet.highlight = TRUE,
  label = TRUE,
  label.size = 3,
  label.color = "white",
  repel = FALSE,
  ncol = NULL,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 1,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  label.box = TRUE,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = FALSE
)

p1 + p2
```




# Searching for features exhibiting spatial patterning in the absence of pre-annotation

Using two methods

# Moran's I Algorithm

Hide

```
GBM <- FindSpatiallyVariableFeatures(GBM,
                                     features = VariableFeatures(GBM)[1:1000],
                                     selection.method = "moransi")
```

```
|                                                                      | 0 % ~calculating
|+                                                                     | 1 % ~11m 13s
|+                                                                     | 2 % ~09m 48s
|++                                                                    | 3 % ~09m 02s
|++                                                                    | 4 % ~08m 39s
|+++                                                                   | 5 % ~08m 23s
|+++                                                                   | 6 % ~08m 13s
|++++                                                                  | 7 % ~08m 01s
|++++                                                                  | 8 % ~07m 51s
|+++++                                                                 | 9 % ~07m 43s
|+++++                                                                 | 10% ~07m 36s
|++++++                                                                | 11% ~07m 35s
|++++++                                                                | 12% ~07m 31s
|+++++++                                                               | 13% ~07m 23s
|+++++++                                                               | 14% ~07m 17s
|++++++++                                                              | 15% ~07m 11s
|++++++++                                                              | 16% ~07m 08s
|+++++++++                                                             | 17% ~07m 04s
|+++++++++                                                             | 18% ~07m 03s
|++++++++++                                                            | 19% ~06m 56s
|++++++++++                                                            | 20% ~06m 50s
|+++++++++++                                                           | 21% ~06m 45s
|+++++++++++                                                           | 22% ~06m 39s
|++++++++++++                                                          | 23% ~06m 34s
|++++++++++++                                                          | 24% ~06m 30s
|+++++++++++++                                                         | 25% ~06m 26s
|+++++++++++++                                                         | 26% ~06m 21s
|++++++++++++++                                                        | 27% ~06m 16s
|++++++++++++++                                                        | 28% ~06m 15s
|+++++++++++++++                                                       | 29% ~06m 11s
|+++++++++++++++                                                       | 30% ~06m 06s
|++++++++++++++++                                                      | 31% ~06m 01s
|++++++++++++++++                                                      | 32% ~05m 55s
|+++++++++++++++++                                                     | 33% ~05m 49s
|+++++++++++++++++                                                     | 34% ~05m 43s
|++++++++++++++++++                                                    | 35% ~05m 38s
|++++++++++++++++++                                                    | 36% ~05m 32s
|+++++++++++++++++++                                                   | 37% ~05m 27s
|+++++++++++++++++++                                                   | 38% ~05m 22s
|++++++++++++++++++++                                                  | 39% ~05m 17s
|++++++++++++++++++++                                                  | 40% ~05m 13s
|+++++++++++++++++++++                                                 | 41% ~05m 07s
|+++++++++++++++++++++                                                 | 42% ~05m 01s
|++++++++++++++++++++++                                                | 43% ~04m 56s
|++++++++++++++++++++++                                                | 44% ~04m 50s
|+++++++++++++++++++++++                                               | 45% ~04m 45s
|+++++++++++++++++++++++                                               | 46% ~04m 39s
|++++++++++++++++++++++++                                              | 47% ~04m 34s
|++++++++++++++++++++++++                                              | 48% ~04m 29s
|+++++++++++++++++++++++++                                             | 49% ~04m 23s
|+++++++++++++++++++++++++                                             | 50% ~04m 18s
|++++++++++++++++++++++++++                                            | 51% ~04m 12s
|++++++++++++++++++++++++++                                            | 52% ~04m 07s
|+++++++++++++++++++++++++++                                           | 53% ~04m 02s
```

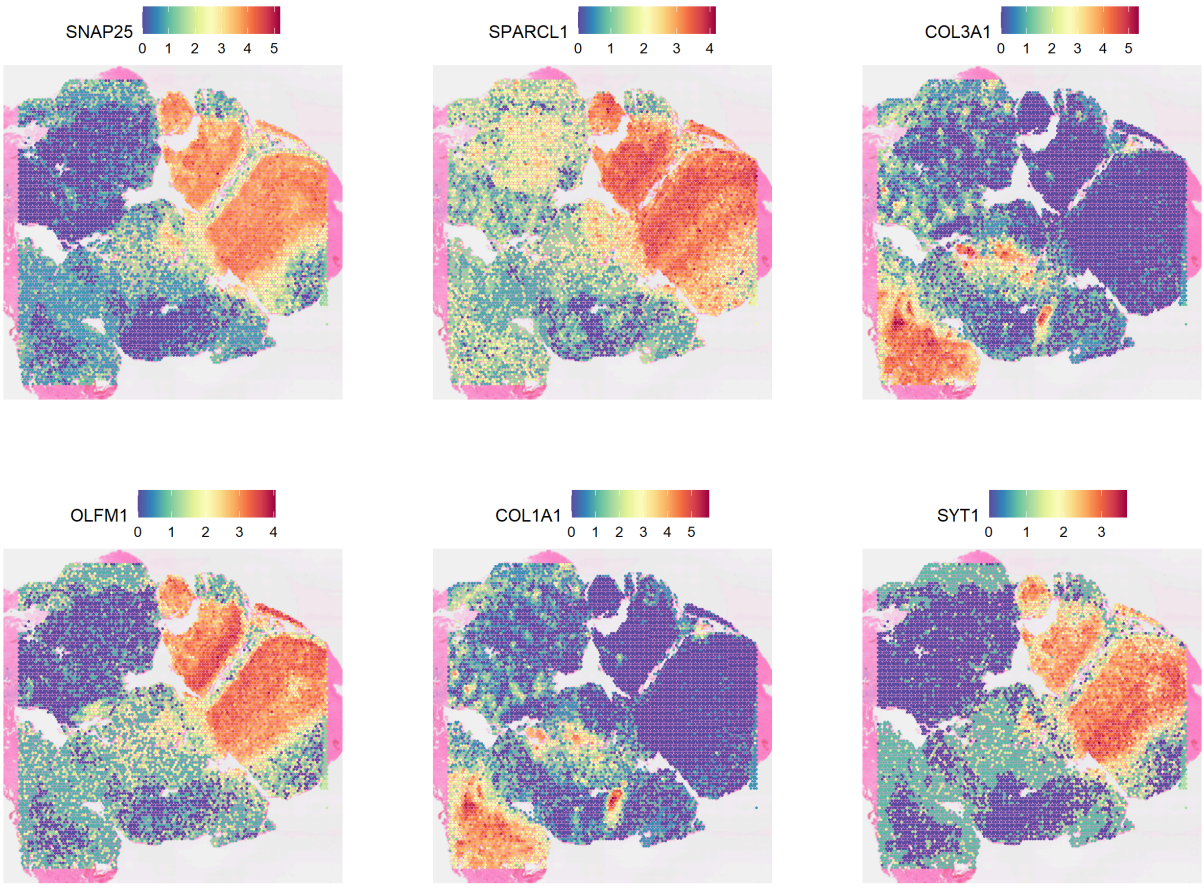
```
|++++++++++++++++++++++++++                          | 54% ~03m 56s
|+++++++++++++++++++++++++++                         | 55% ~03m 51s
|++++++++++++++++++++++++++++                        | 56% ~03m 45s
|++++++++++++++++++++++++++++                        | 57% ~03m 40s
|++++++++++++++++++++++++++++                        | 58% ~03m 34s
|+++++++++++++++++++++++++++++                       | 59% ~03m 29s
|+++++++++++++++++++++++++++++                       | 60% ~03m 23s
|+++++++++++++++++++++++++++++                       | 61% ~03m 18s
|++++++++++++++++++++++++++++++                      | 62% ~03m 13s
|++++++++++++++++++++++++++++++                      | 63% ~03m 07s
|++++++++++++++++++++++++++++++                      | 64% ~03m 02s
|+++++++++++++++++++++++++++++++                     | 65% ~02m 57s
|+++++++++++++++++++++++++++++++                     | 66% ~02m 51s
|+++++++++++++++++++++++++++++++                     | 67% ~02m 46s
|++++++++++++++++++++++++++++++++                    | 68% ~02m 41s
|++++++++++++++++++++++++++++++++                    | 69% ~02m 36s
|++++++++++++++++++++++++++++++++                    | 70% ~02m 31s
|+++++++++++++++++++++++++++++++++                   | 71% ~02m 26s
|+++++++++++++++++++++++++++++++++                   | 72% ~02m 21s
|+++++++++++++++++++++++++++++++++                   | 73% ~02m 16s
|++++++++++++++++++++++++++++++++++                  | 74% ~02m 10s
|++++++++++++++++++++++++++++++++++                  | 75% ~02m 05s
|++++++++++++++++++++++++++++++++++                  | 76% ~02m 00s
|+++++++++++++++++++++++++++++++++++                 | 77% ~01m 55s
|+++++++++++++++++++++++++++++++++++                 | 78% ~01m 50s
|+++++++++++++++++++++++++++++++++++                 | 79% ~01m 45s
|++++++++++++++++++++++++++++++++++++                | 80% ~01m 40s
|++++++++++++++++++++++++++++++++++++                | 81% ~01m 35s
|++++++++++++++++++++++++++++++++++++                | 82% ~01m 30s
|+++++++++++++++++++++++++++++++++++++               | 83% ~01m 25s
|+++++++++++++++++++++++++++++++++++++               | 84% ~01m 20s
|+++++++++++++++++++++++++++++++++++++               | 85% ~01m 15s
|++++++++++++++++++++++++++++++++++++++              | 86% ~01m 10s
|++++++++++++++++++++++++++++++++++++++              | 87% ~01m 05s
|++++++++++++++++++++++++++++++++++++++              | 88% ~01m 00s
|+++++++++++++++++++++++++++++++++++++++             | 89% ~55s
|+++++++++++++++++++++++++++++++++++++++             | 90% ~50s
|+++++++++++++++++++++++++++++++++++++++             | 91% ~45s
|++++++++++++++++++++++++++++++++++++++++            | 92% ~40s
|++++++++++++++++++++++++++++++++++++++++            | 93% ~35s
|++++++++++++++++++++++++++++++++++++++++            | 94% ~30s
|+++++++++++++++++++++++++++++++++++++++++           | 95% ~25s
|+++++++++++++++++++++++++++++++++++++++++           | 96% ~20s
|++++++++++++++++++++++++++++++++++++++++++          | 97% ~15s
|++++++++++++++++++++++++++++++++++++++++++          | 98% ~10s
|++++++++++++++++++++++++++++++++++++++++++          | 99% ~05s
|++++++++++++++++++++++++++++++++++++++++++          | 100% elapsed=08m 22s
```

Hide

```
# With data

top.features <- head(SpatiallyVariableFeatures(GBM, method = "moransi"), 6)

SpatialFeaturePlot(
  GBM,
  features = top.features,
  images = NULL,
  crop = TRUE,
  slot = "data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = 3,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(2, 2),
  image.alpha = 2,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
) &
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))
```




```
```

Interestingly, Clusters 15, 11, 13, 6, 3, 16, 12 have a common base of overly expressed group of genes (could they be actually way more similar than expected and therefore PCA is overclustering?)

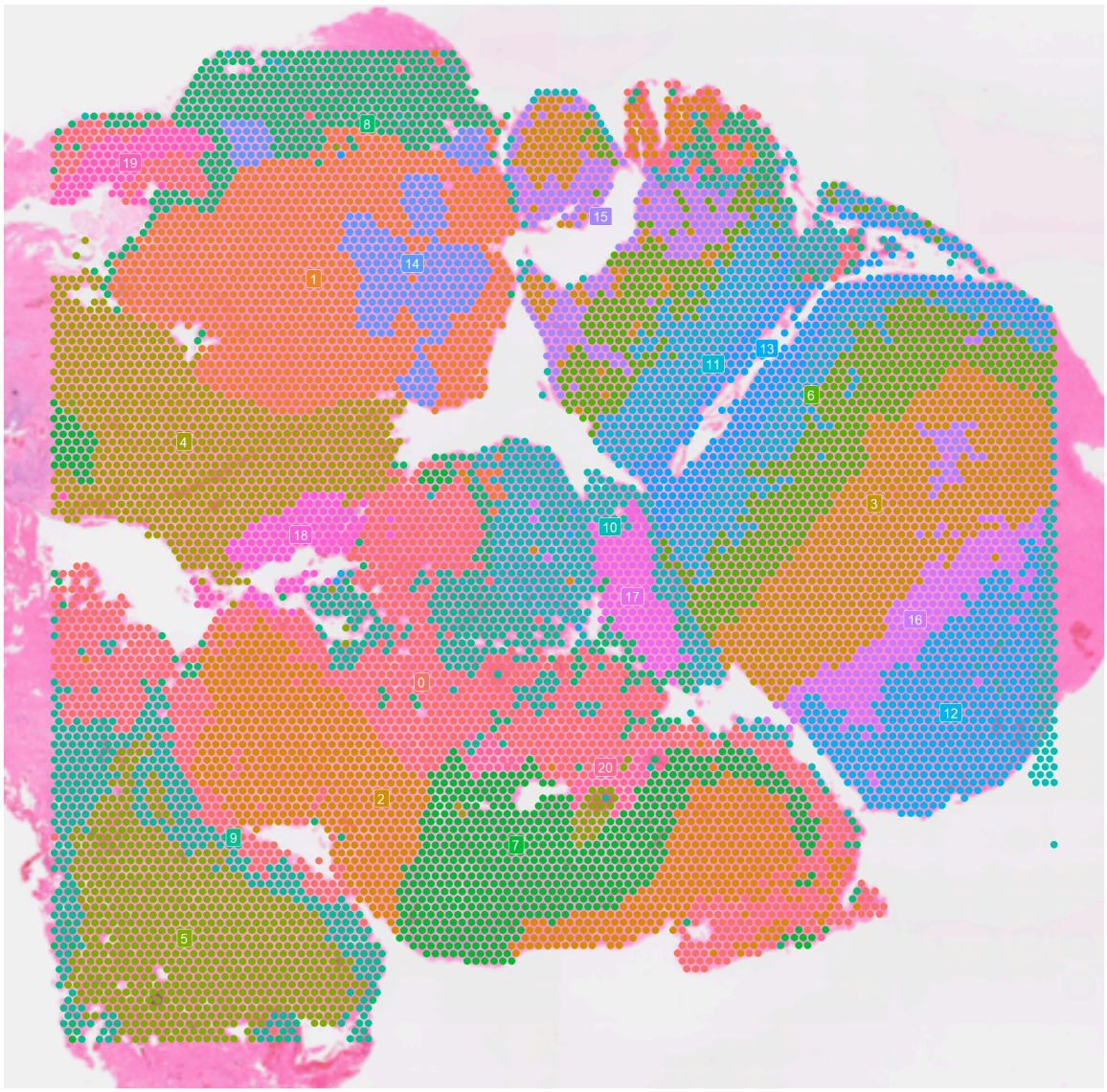
```
```

Hide

```
# With scale data

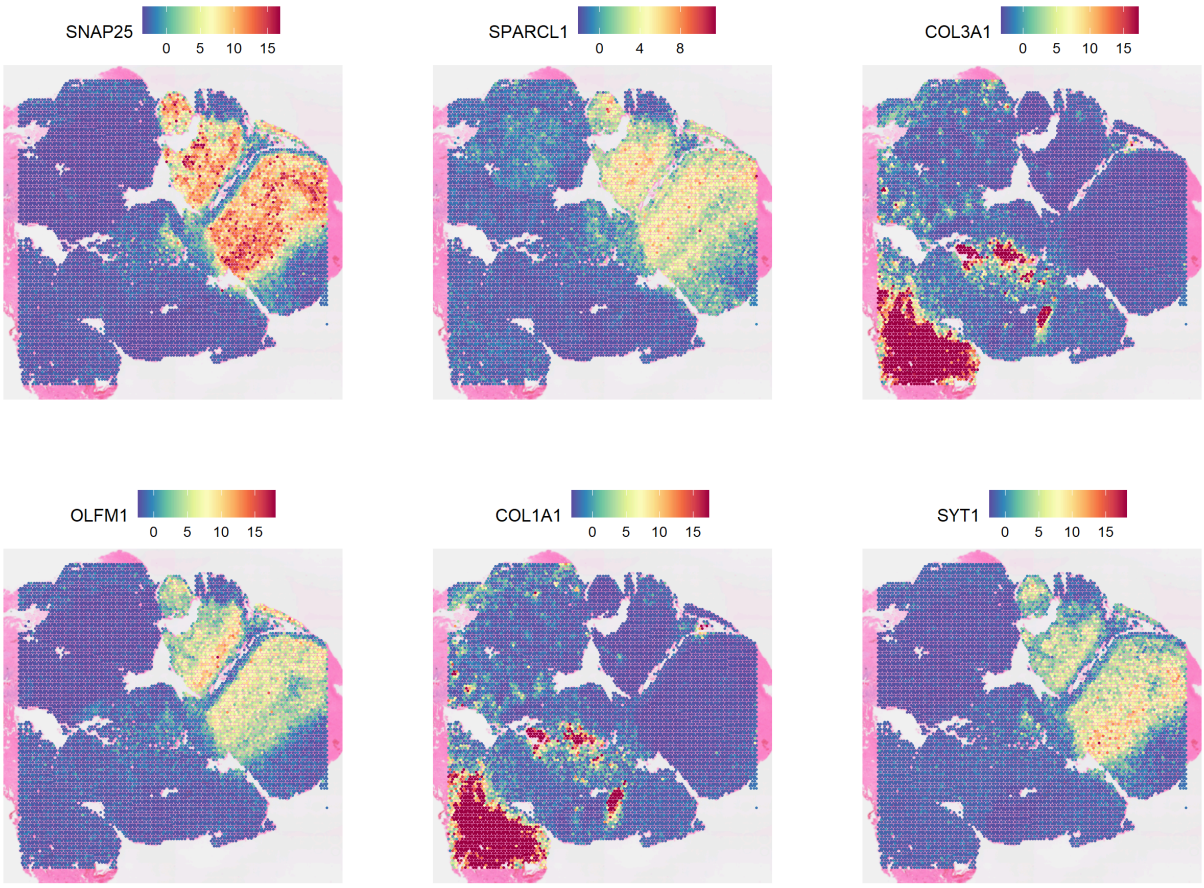
top.features <- head(SpatiallyVariableFeatures(GBM, method = "moransi"), 6)

SpatialDimPlot(
  GBM,
  group.by = NULL,
  images = NULL,
  cols = NULL,
  crop = TRUE,
  cells.highlight = NULL,
  cols.highlight = c("#DE2D26", "grey50"),
  facet.highlight = TRUE,
  label = TRUE,
  label.size = 3,
  label.color = "white",
  repel = FALSE,
  ncol = NULL,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 1,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  label.box = TRUE,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = FALSE
) +
  NoLegend()
```




Hide

```
SpatialFeaturePlot(
  GBM,
  features = top.features,
  images = NULL,
  crop = TRUE,
  slot = "scale.data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = 3,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(2, 2),
  image.alpha = 2,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
) &
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))
```




# Mark Variogram Algorithm

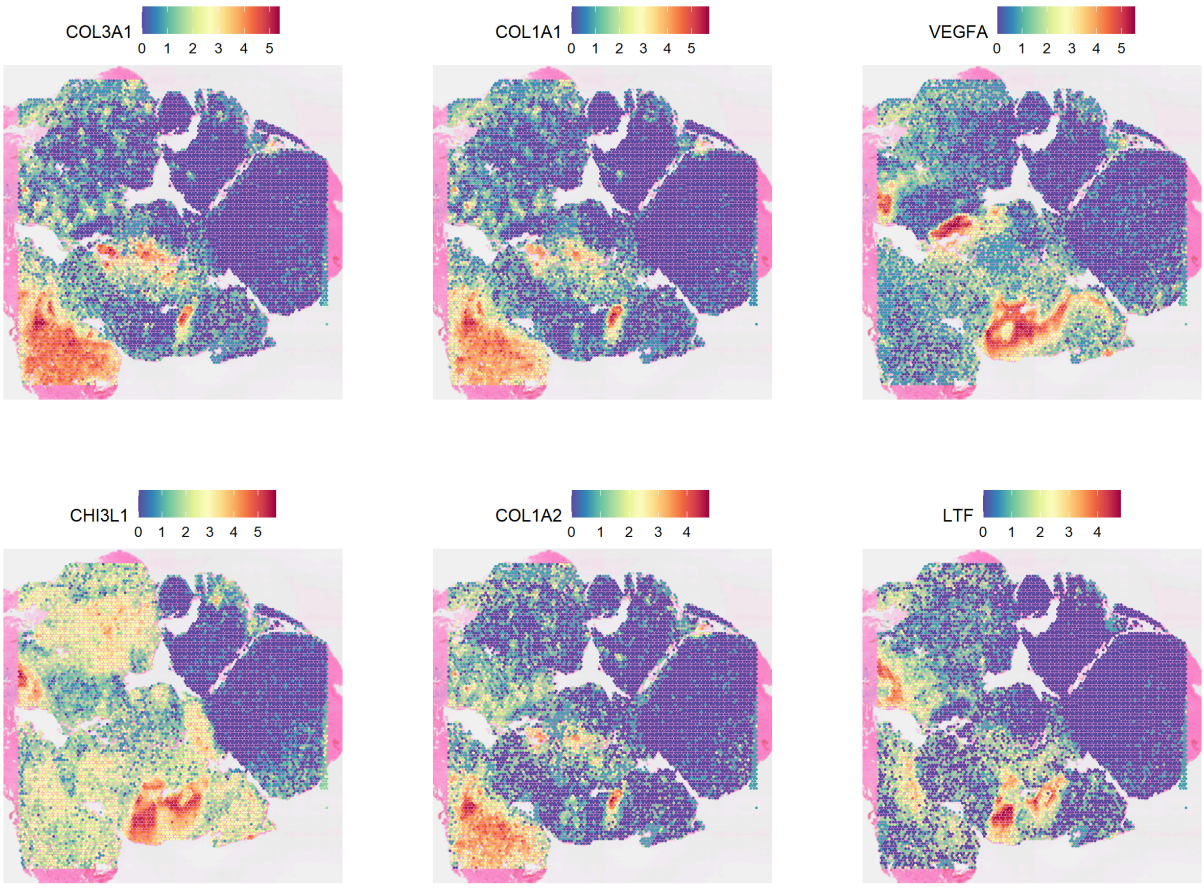
Hide

```
GBM <- FindSpatiallyVariableFeatures(GBM,
                                     features = VariableFeatures(GBM)[1:10],
                                     selection.method = "markvariogram")
```

Hide

```
# With data
top.features <- head(SpatiallyVariableFeatures(GBM, method = "markvariogram"), 6)

SpatialFeaturePlot(
  GBM,
  features = top.features,
  images = NULL,
  crop = TRUE,
  slot = "data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = 3,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(2, 2),
  image.alpha = 2,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
) &
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))
```



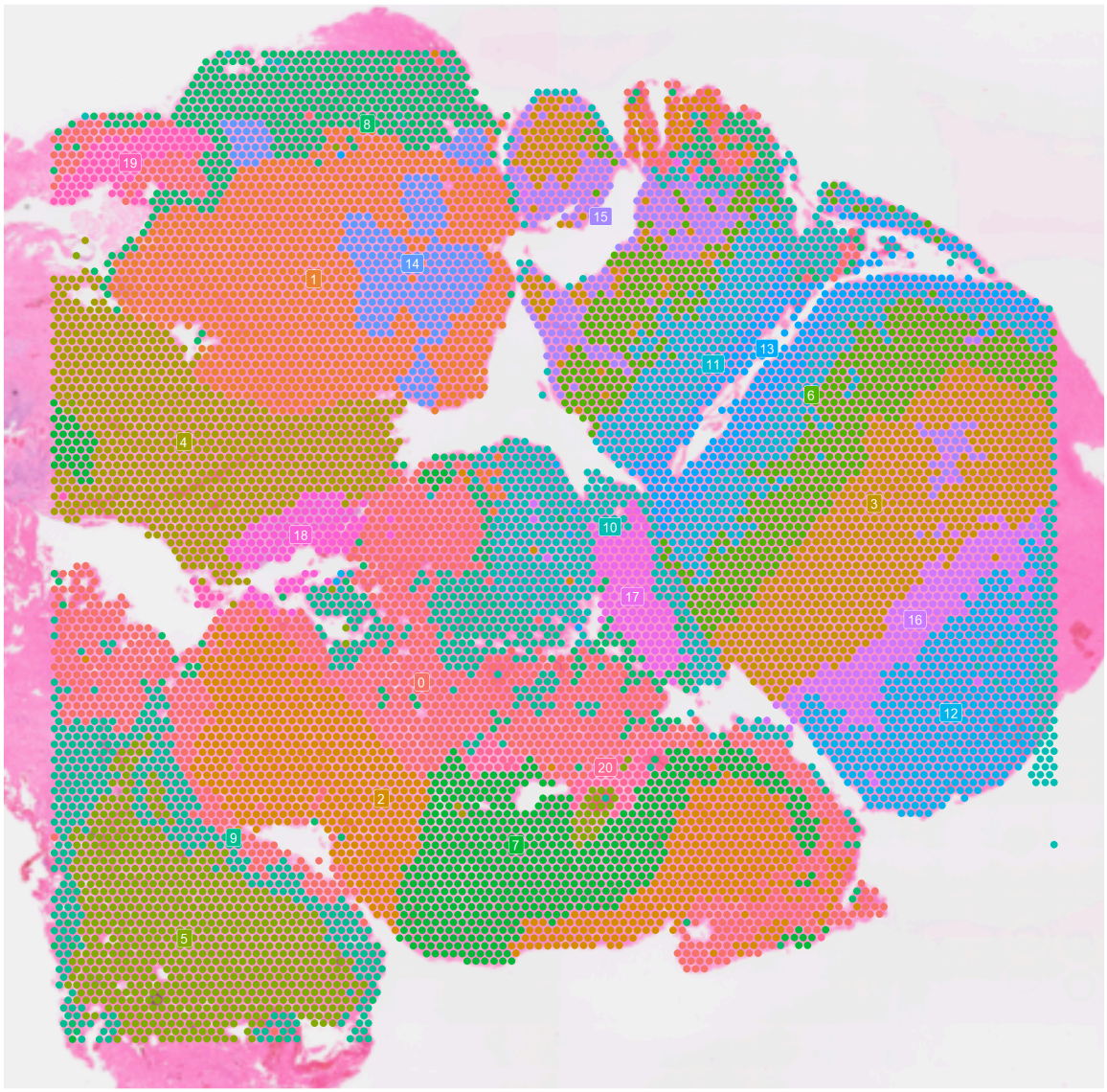

The patterns of Spatially Variable Features most certainly changed between algorythms, I will be moving forward with

'COL' type genes seem to be common to a certain specific region of the brain with both algorythms.

Hide

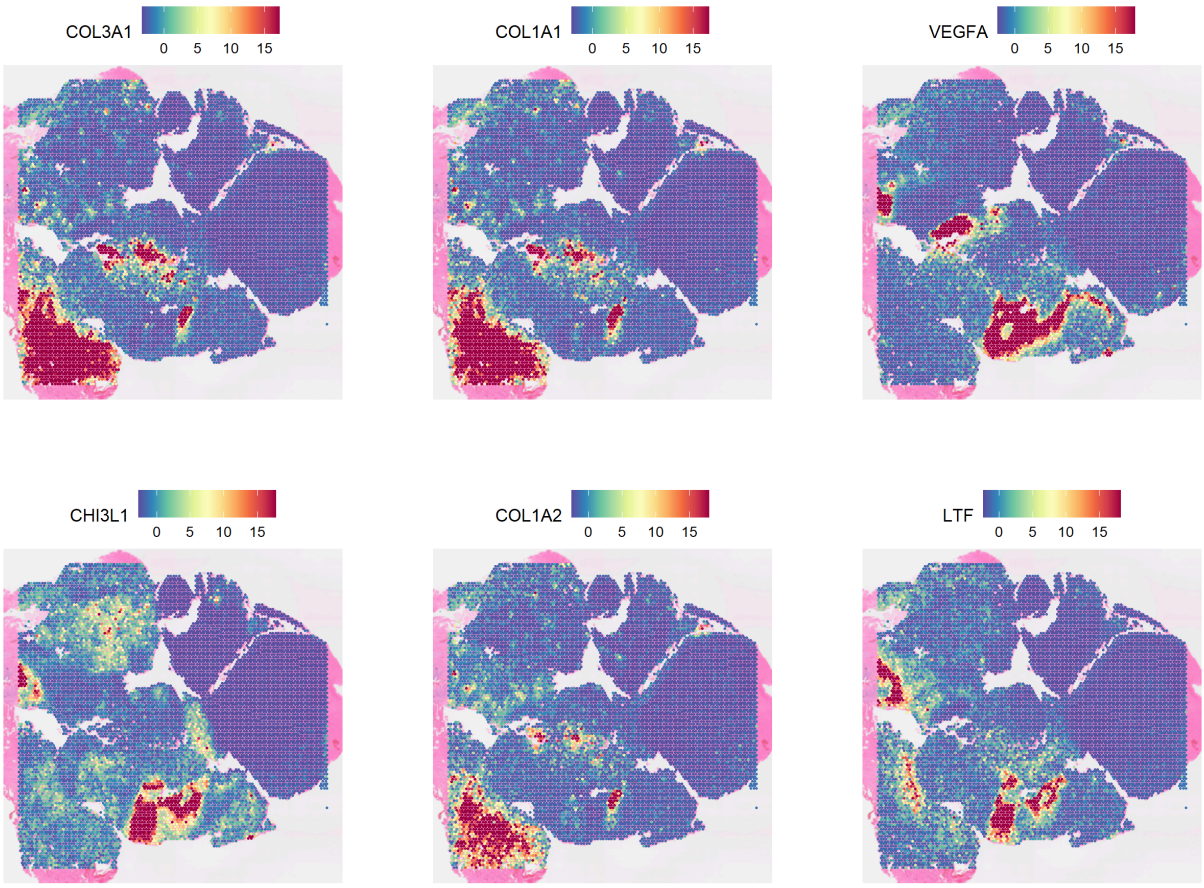
```
# With scale data
top.features <- head(SpatiallyVariableFeatures(GBM, method = "markvariogram"), 6)

SpatialDimPlot(
  GBM,
  group.by = NULL,
  images = NULL,
  cols = NULL,
  crop = TRUE,
  cells.highlight = NULL,
  cols.highlight = c("#DE2D26", "grey50"),
  facet.highlight = TRUE,
  label = TRUE,
  label.size = 3,
  label.color = "white",
  repel = FALSE,
  ncol = NULL,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 1,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  label.box = TRUE,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = FALSE
) +
  NoLegend()
```




Hide

```
SpatialFeaturePlot(
  GBM,
  features = top.features,
  images = NULL,
  crop = TRUE,
  slot = "scale.data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = 3,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(2, 2),
  image.alpha = 2,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
) &
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))
```




# Inspecting clusters and finding their TOP markers to study patterns

Hide

```
table(Idents(GBM))
```

```
   0    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15   16   17   1
8   19   20
1262  969  902  795  757  725  677  649  616  571  557  482  478  307  272  255  214  124  11
2   91   63
```

Hide

```
markers <- FindAllMarkers(
  GBM,
  only.pos = TRUE,
  min.pct = 0.25,
  logfc.threshold = 0.25
)
```

# 10 Top Genes per cluster

Hide

```
library(dplyr)

top10 <- markers %>%
  group_by(cluster) %>%
  slice_max(
    order_by = avg_log2FC,
    n = 10
  )

top10
```

| p_val <dbl> | avg_log2FC <dbl> | pct.1 <dbl> | pct.2 <dbl> | p_val_adj <dbl> | cluster <fctr> | gene <chr> |
|---|---|---|---|---|---|---|
| 6.937356e-280 | 2.0979328 | 0.666 | 0.246 | 1.253650e-275 | 0 | IL26 |
| 1.251727e-243 | 2.0844889 | 0.615 | 0.229 | 2.261996e-239 | 0 | IL22 |
| 2.726333e-118 | 1.5930694 | 0.563 | 0.333 | 4.926756e-114 | 0 | NUP107 |
| 1.885771e-40 | 1.4564387 | 0.254 | 0.130 | 3.407777e-36 | 0 | MUC12 |
| 6.712439e-111 | 1.4468966 | 0.605 | 0.407 | 1.213005e-106 | 0 | SZRD1 |
| 1.145808e-35 | 1.3234250 | 0.268 | 0.147 | 2.070589e-31 | 0 | FLG |
| 7.053281e-239 | 1.2664401 | 0.999 | 0.833 | 1.274598e-234 | 0 | CAND1 |
| 2.711153e-89 | 1.2374792 | 0.539 | 0.309 | 4.899325e-85 | 0 | ELOA3 |
| 7.295363e-97 | 1.2373416 | 0.575 | 0.345 | 1.318345e-92 | 0 | FAM90A1 |
| 2.490170e-164 | 1.2184393 | 0.805 | 0.438 | 4.499986e-160 | 0 | E2F3 |

1-10 of 210 rows Previous **1** 2 3 4 5 6 … 21 Next

Hide

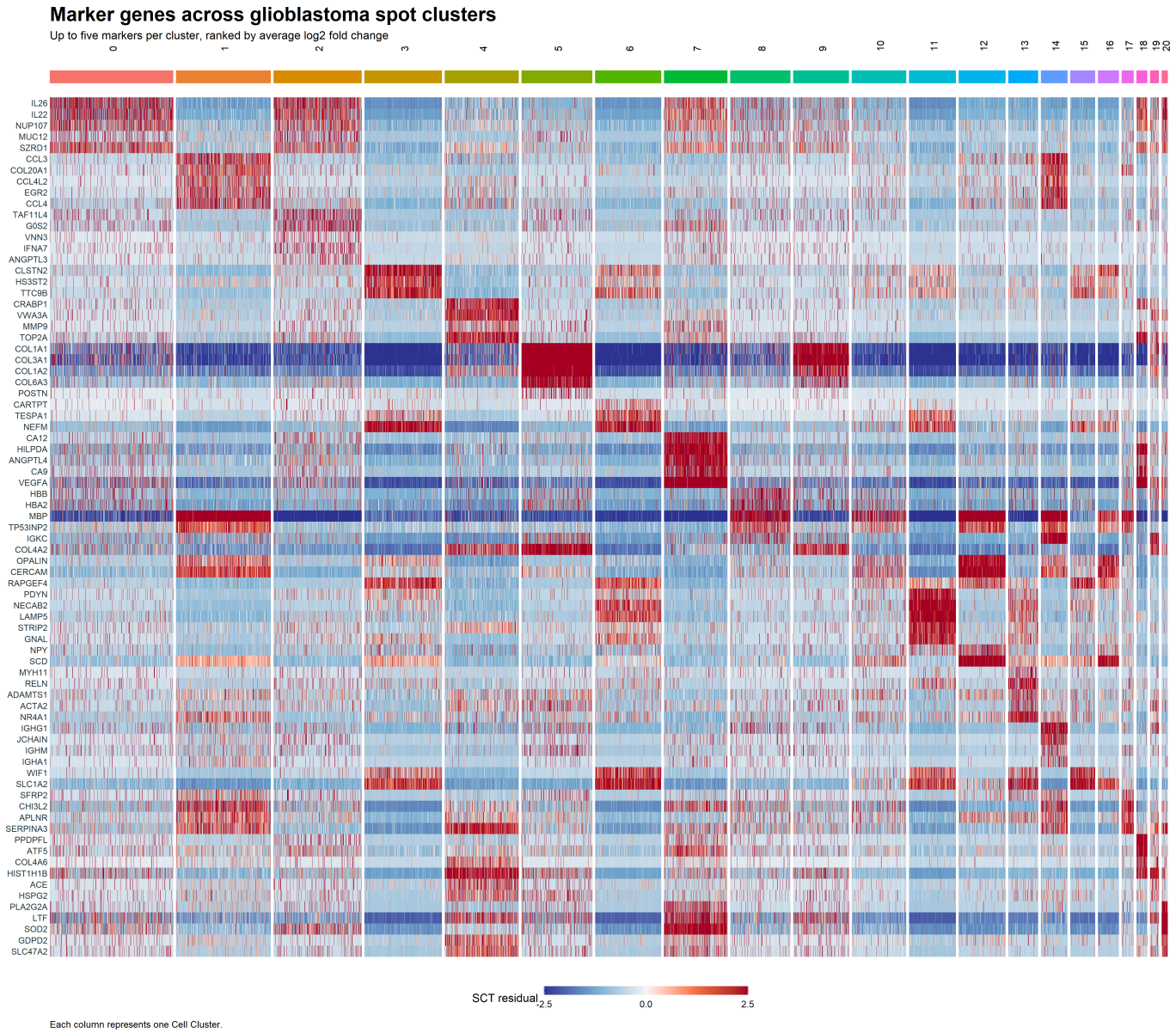
```
top5 <- top10 %>%
  group_by(cluster) %>%
  slice_max(order_by = avg_log2FC, n = 5, with_ties = FALSE) %>%
  ungroup()

residual_limit <- 2.5

p_heatmap <- DoHeatmap(
  GBM,
  features = unique(top5$gene),
  assay = "SCT",
  slot = "scale.data",
  group.by = "seurat_clusters",
  disp.min = -residual_limit,
  disp.max = residual_limit,
  angle = 90,
  hjust = 0,
  vjust = 0.5,
  size = 3.4,
  group.bar.height = 0.015,
  combine = FALSE
)[[1]] +
  scale_fill_gradientn(
    colours = c("#313695", "#74ADD1", "#F7F7F7", "#F46D43", "#A50026"),
    limits = c(-residual_limit, residual_limit),
    breaks = c(-residual_limit, 0, residual_limit),
    na.value = "white",
    name = "SCT residual"
  ) +
  labs(
    title = "Marker genes across glioblastoma spot clusters",
    subtitle = "Up to five markers per cluster, ranked by average log2 fold change",
    caption = "Each column represents one Cell Cluster."
  ) +
  theme(
    axis.text.y = element_text(size = 8.5, colour = "#263238"),
    legend.position = "bottom",
    legend.key.width = grid::unit(14, "mm"),
    legend.key.height = grid::unit(3, "mm"),
    plot.title = element_text(size = 19, face = "bold"),
    plot.subtitle = element_text(size = 11, margin = margin(b = 10)),
    plot.caption = element_text(size = 9, hjust = 0),
    plot.margin = margin(8, 12, 6, 8)
  )

p_heatmap_clus <- p_heatmap + ggplot2::guides(colour = "none")

print(p_heatmap_clus)
```




Hide

```
ggplot2::ggsave(
  filename = "GBM_Top5_Marker_Heatmap.png",
  plot = p_heatmap_clus,
  width = 16,
  height = 14,
  units = "in",
  dpi = 200,
  bg = "white"
)
```

# Possible correlation

# Identification of Spatially Variable Features using Seurat

Perform differential expression based on pre-annotated anatomical regions within the tissue, which may be determined either from unsupervised clustering or prior knowledge.

Cluster 5 and 9 have ‘COL’ type genes overly expressed when compared to all other clusters

Hide

```
avg_sct_residuals <- as.matrix(
  AverageExpression(
    GBM,
    assays = "SCT",
    group.by = "seurat_clusters",
    layer = "scale.data"
  )$SCT
)

rho <- cor(avg_sct_residuals, method = "spearman")
cor_5_9 <- rho["g5", "g9"]

matches_5 <- sort(
  rho["g5", colnames(rho) != "g5"],
  decreasing = TRUE
)

data.frame(
  rank = seq_along(matches_5),
  cluster = sub("^g", "", names(matches_5)),
  spearman_rho = round(unname(matches_5), 3)
)
```

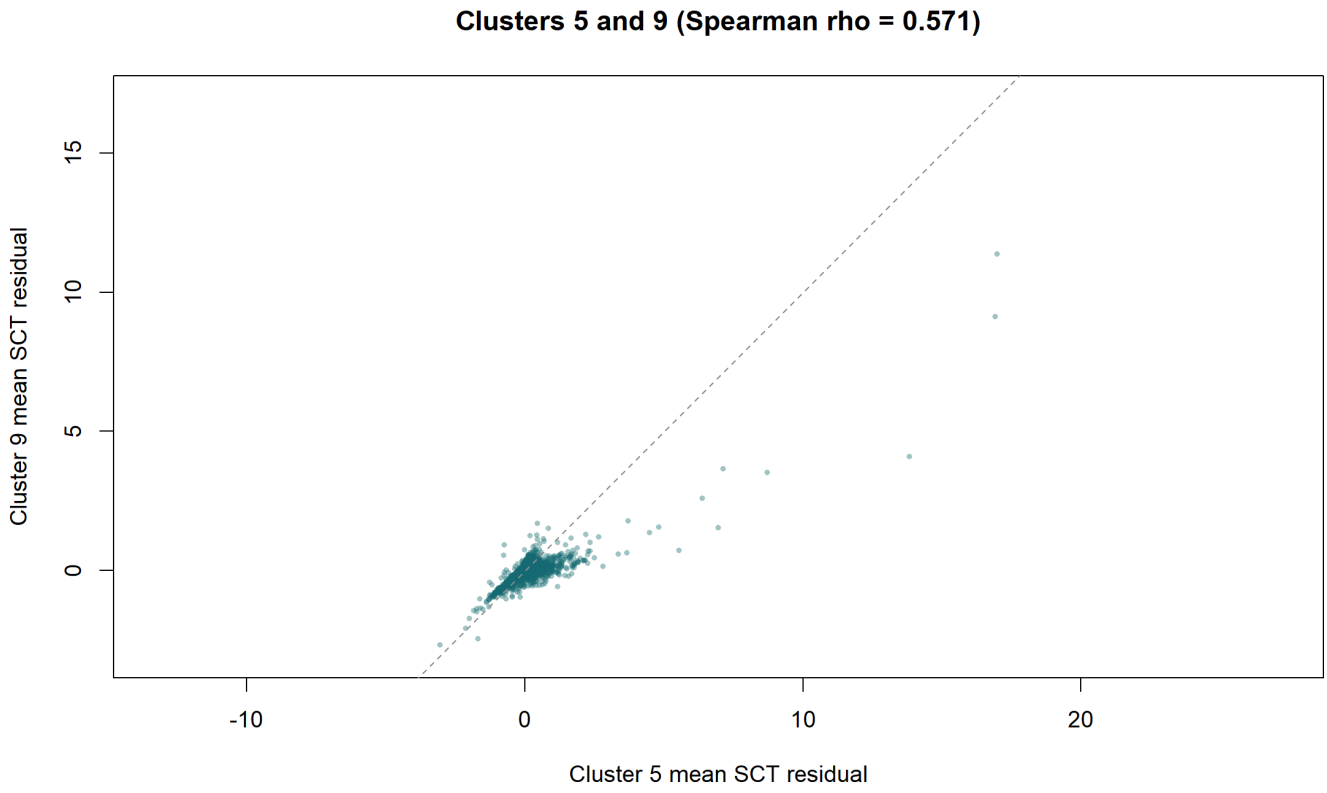
| rank | cluster | spearman_rho |
|---|---|---|
| <int> | <chr> | <dbl> |
| 1 | 9 | 0.571 |
| 2 | 19 | 0.540 |
| 3 | 4 | 0.483 |
| 4 | 14 | 0.447 |
| 5 | 8 | 0.435 |
| 6 | 1 | 0.420 |
| 7 | 20 | 0.342 |
| 8 | 7 | 0.335 |
| 9 | 0 | 0.283 |
| 10 | 2 | 0.240 |

1-10 of 20 rows Previous **1** 2 Next

Hide

```
pair <- avg_sct_residuals[, c("g5", "g9")]
limits <- range(pair)

plot(
  pair[, "g5"], pair[, "g9"],
  xlim = limits, ylim = limits, asp = 1,
  pch = 16, cex = 0.55,
  col = adjustcolor("#176C72", alpha.f = 0.4),
  xlab = "Cluster 5 mean SCT residual",
  ylab = "Cluster 9 mean SCT residual",
  main = sprintf("Clusters 5 and 9 (Spearman rho = %.3f)", cor_5_9)
)
abline(a = 0, b = 1, lty = 2, col = "grey55")
```




# Mapping the clusters and gene programms

This highlights clusters 5 (Collagen), 9 (Collagen) and 7 (Hypoxia) on the same H&E image.

Hide

```
focus <- c("5", "9", "7")
cluster <- as.character(GBM$seurat_clusters)
GBM$focus_region <- factor(
  ifelse(cluster %in% focus, cluster, "Other"),
  levels = c(focus, "Other")
)

p_clusters <- SpatialDimPlot(
  GBM,
  group.by = "focus_region",
  pt.size.factor = 2.8,
  alpha = 1,
  image.alpha = 0.55,
  cols = c(
    "5" = "#D55E00",
    "9" = "#009E73",
    "7" = "#0072B2",
    "Other" = "grey88"
  ),
  combine = FALSE
)[[1]]

panels <- list(
  Collagen = c("COL1A1", "COL1A2", "COL3A1", "COL6A3"),
  GBM_markers = c("BCAN", "PTPRZ1", "DDR1"),
  Hypoxia = c("CA12", "HILPDA", "ANGPTL4", "CA9", "VEGFA"),
  MES1_markers = c("CHI3L1", "CD44", "ANXA2"),
  MES2_markers = c("ADM", "LDHA", "DDIT3")
)

panels <- lapply(panels, intersect, y = rownames(GBM[["SCT"]]))
print(panels)
```

```
$Collagen
[1] "COL1A1" "COL1A2" "COL3A1" "COL6A3"

$GBM_markers
[1] "BCAN"   "PTPRZ1" "DDR1"

$Hypoxia
[1] "CA12"    "HILPDA"  "ANGPTL4" "CA9"     "VEGFA"

$MES1_markers
[1] "CHI3L1" "CD44"   "ANXA2"

$MES2_markers
[1] "ADM"   "DDIT3"
```

Hide

```
GBM <- AddModuleScore(
  GBM, features = panels, assay = "SCT", name = "Niche", seed = 42
)
GBM$Collagen <- GBM$Niche1
GBM$GBM_markers <- GBM$Niche2
GBM$Hypoxia <- GBM$Niche3
GBM$MES1_markers <- GBM$Niche4
GBM$MES2_markers <- GBM$Niche5

score_maps <- SpatialFeaturePlot(
  GBM,
  features = c(
    "Collagen", "Hypoxia",
    "MES1_markers", "MES2_markers",
    "GBM_markers"
  ),
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 0.55,
  keep.scale = "feature",
  combine = FALSE
)
```

Hide

```
## Not a must, just making the graphs more readble
p_clusters <- p_clusters +
  ggplot2::guides(
    fill = ggplot2::guide_legend(
      override.aes = list(size = 6, alpha = 1)
    )
  )

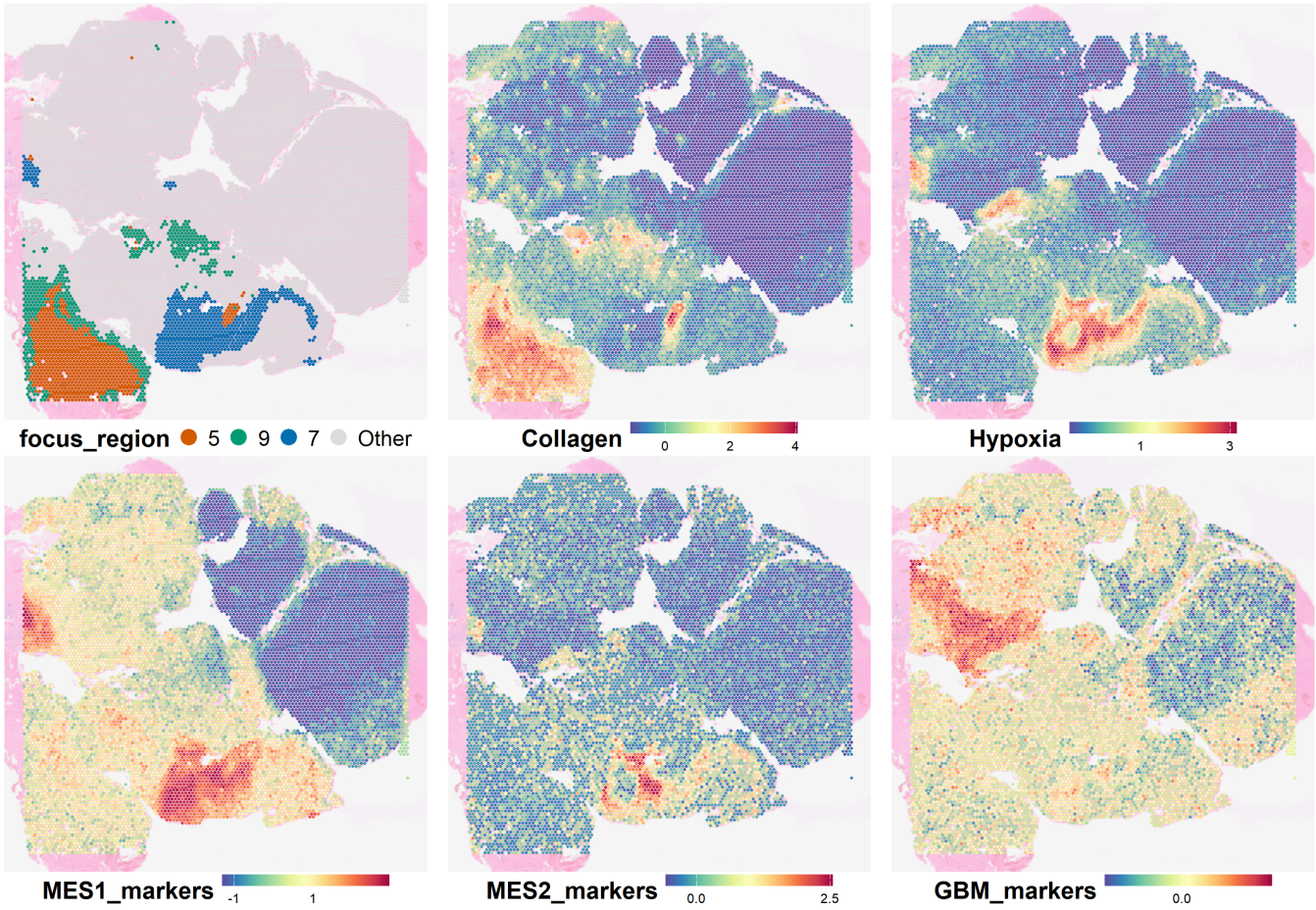
score_maps <- lapply(score_maps, function(p) {
  fill_scale <- p$scales$get_scales("fill")
  fill_scale$breaks <- scales::breaks_extended(n = 3)

  p + ggplot2::guides(
    fill = ggplot2::guide_colourbar(
      direction = "horizontal",
      theme = ggplot2::theme(
        legend.key.width = grid::unit(44, "mm"),
        legend.key.height = grid::unit(3, "mm"),
        legend.text = ggplot2::element_text(size = 10)
      )
    )
  )
})

## Plotting

p_spatial <- wrap_plots(
  c(list(p_clusters), score_maps),
  ncol = 3,
  nrow = 2,
  guides = "keep"
) & ggplot2::theme(
  plot.margin = ggplot2::margin(t = 1, r = 8, b = 1, l = 8, unit = "pt"),
  legend.position = "bottom",
  legend.box.spacing = grid::unit(2, "pt"),
  legend.margin = ggplot2::margin(0, 0, 0, 0),
  legend.text = ggplot2::element_text(size = 16),
  legend.title = ggplot2::element_text(size = 18, face = "bold")
)

p_spatial
```




# Comparing Clusters highly expression MBP

Hide

```
avg_expr <- AverageExpression(
  GBM,
  assays = "SCT",
  group.by = "seurat_clusters",
  layer = "scale.data"
)$SCT

clusters_to_compare <- c(
  "g1", "g10", "g14", "g17",
  "g8", "g12", "g16"
)

avg_expr_subset <- as.matrix(
  avg_expr[, clusters_to_compare, drop = FALSE]
)

cor_matrix <- cor(
  avg_expr_subset,
  method = "spearman"
)

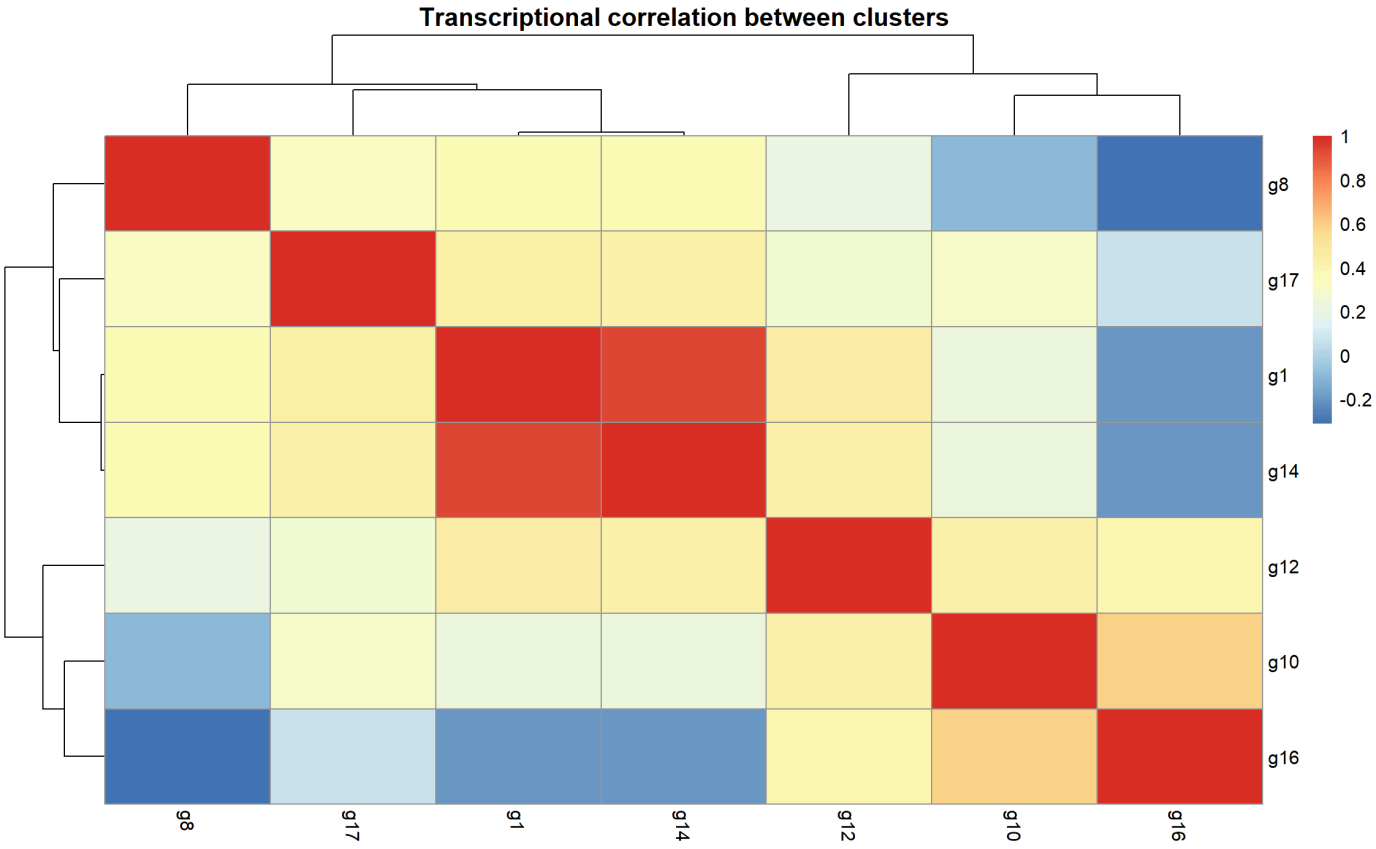
round(cor_matrix, 3)
```

```
        g1    g10    g14   g17     g8   g12    g16
g1   1.000  0.224  0.944 0.440  0.374 0.457 -0.194
g10  0.224  1.000  0.229 0.299 -0.090 0.431  0.587
g14  0.944  0.229  1.000 0.424  0.376 0.434 -0.202
g17  0.440  0.299  0.424 1.000  0.320 0.277  0.065
g8   0.374 -0.090  0.376 0.320  1.000 0.210 -0.312
g12  0.457  0.431  0.434 0.277  0.210 1.000  0.407
g16 -0.194  0.587 -0.202 0.065 -0.312 0.407  1.000
```

Hide

```
library(pheatmap)

pheatmap(
  cor_matrix,
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  main = "Transcriptional correlation between clusters"
)
```




Hide

```
cor_pairs <- as.data.frame(as.table(cor_matrix))

cor_pairs <- cor_pairs %>%
  filter(Var1 != Var2) %>%
  mutate(
    pair = paste(
      pmin(as.character(Var1), as.character(Var2)),
      pmax(as.character(Var1), as.character(Var2)),
      sep = " vs "
    )
  ) %>%
  distinct(pair, .keep_all = TRUE) %>%
  arrange(desc(Freq))

cor_pairs
```

| Var1 <fctr> | Var2 <fctr> | Freq <dbl> | pair <chr> |
|---|---|---|---|
| g14 | g1 | 0.94376386 | g1 vs g14 |
| g16 | g10 | 0.58682826 | g10 vs g16 |
| g12 | g1 | 0.45725566 | g1 vs g12 |
| g17 | g1 | 0.43991782 | g1 vs g17 |
| g12 | g14 | 0.43404108 | g12 vs g14 |
| g12 | g10 | 0.43060405 | g10 vs g12 |
| g17 | g14 | 0.42397366 | g14 vs g17 |
| g16 | g12 | 0.40749440 | g12 vs g16 |
| g8 | g14 | 0.37597663 | g14 vs g8 |
| g8 | g1 | 0.37352477 | g1 vs g8 |

1-10 of 21 rows Previous **1** 2 3 Next

Hide

```
library(Seurat)
library(dplyr)
library(ggplot2)
library(patchwork)

gbm_myelin <- GBM
DefaultAssay(gbm_myelin) <- "SCT"

support_genes <- c("PLP1", "MOG", "MOBP")
genes <- c("MBP", support_genes)

missing_genes <- setdiff(genes, rownames(gbm_myelin[["SCT"]]))
if (length(missing_genes) > 0) {
  stop("Genes missing from the SCT assay: ", paste(missing_genes, collapse = ", "))
}

cluster_ids <- as.character(gbm_myelin$seurat_clusters)
gbm_myelin$myelin_region <- factor(
  ifelse(cluster_ids %in% c("1", "14"), paste("Cluster", cluster_ids), "Other spots"),
  levels = c("Cluster 1", "Cluster 14", "Other spots")
)

gbm_myelin <- AddModuleScore(
  gbm_myelin,
  features = list(support_genes),
  assay = "SCT",
  name = "MyelinCheck",
  seed = 42
)

image_id <- Images(gbm_myelin)[1]

p_clusters <- SpatialDimPlot(
  gbm_myelin,
  group.by = "myelin_region",
  images = image_id,
  image.scale = "lowres",
  cols = c(
    "Cluster 1" = "#D55E00",
    "Cluster 14" = "#0072B2",
    "Other spots" = "#D9D9D9"
  ),
  pt.size.factor = 2.5,
  image.alpha = 0.55,
  combine = FALSE
)[[1]] + ggtitle("Clusters 1 and 14")

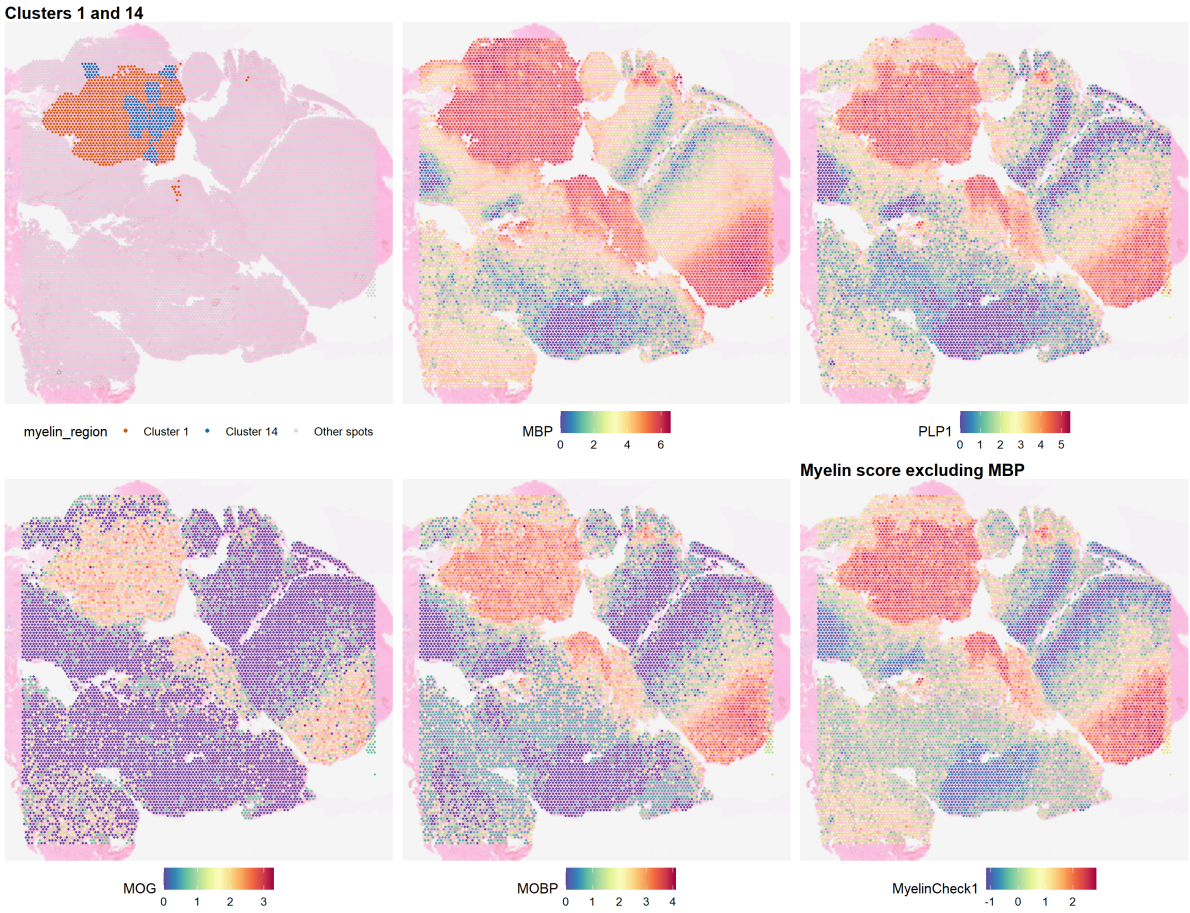
marker_maps <- SpatialFeaturePlot(
  gbm_myelin,
  features = c(genes, "MyelinCheck1"),
  slot = "data",
  images = image_id,
  image.scale = "lowres",
  pt.size.factor = 2.5,
  image.alpha = 0.55,
```

```
  keep.scale = "feature",
  combine = FALSE
)

score_panel <- length(marker_maps)
marker_maps[[score_panel]] <- marker_maps[[score_panel]] +
  ggtitle("Myelin score excluding MBP")

p_myelin <- wrap_plots(c(list(p_clusters), marker_maps), ncol = 3) &
  theme(
    plot.margin = margin(2, 4, 2, 4),
    legend.position = "bottom",
    plot.title = element_text(size = 13, face = "bold")
  )

p_myelin
```




Hide

```
spot_data <- FetchData(
  gbm_myelin,
  vars = c("myelin_region", genes, "MyelinCheck1"),
  layer = "data"
)

spot_data %>%
  group_by(myelin_region) %>%
  summarise(
    spots = n(),
    across(all_of(genes), mean, .names = "mean_{.col}"),
    median_myelin_score = median(MyelinCheck1),
    .groups = "drop"
  )
```

| myelin_region <fctr> | sp… <int> | mean_… <dbl> | mean_P… <dbl> | mean_M… <dbl> | mean_M… <dbl> | median_myelin_score <dbl> |
|---|---|---|---|---|---|---|
| Cluster 1 | 969 | 5.496520 | 4.420018 | 1.8037202 | 2.963050 | 2.1170766 |
| Cluster 14 | 272 | 5.442496 | 4.386883 | 1.7914450 | 2.936575 | 2.1202915 |
| Other spots | 9637 | 3.077577 | 1.917730 | 0.4151402 | 1.046296 | 0.1013383 |

3 rows

Hide

```
mbp_spots <- FetchData(
  gbm_myelin,
  vars = c("seurat_clusters", "MBP", "MyelinCheck1", "nCount_Spatial"),
  layer = "data"
)

focus <- c("1", "14")
mbp_cutoff <- quantile(mbp_spots$MBP, 0.80, names = FALSE)

mbp_spots$cluster <- as.character(mbp_spots$seurat_clusters)
mbp_spots$mbp_high <- mbp_spots$MBP > 0 & mbp_spots$MBP >= mbp_cutoff

mbp_summary <- mbp_spots %>%
  group_by(cluster) %>%
  summarise(
    spots = n(),
    high_spots = sum(mbp_high),
    pct_high = round(100 * mean(mbp_high), 1),
    median_MBP_high = median(MBP[mbp_high]),
    median_myelin_high = median(MyelinCheck1[mbp_high]),
    median_UMIs_high = median(nCount_Spatial[mbp_high]),
    .groups = "drop"
  ) %>%
  arrange(desc(pct_high))

print(mbp_summary, n = Inf)
```

```
# A tibble: 21 × 7
   cluster spots high_spots pct_high median_MBP_high median_myelin_high median_UMIs_high
   <chr>   <int>      <int>    <dbl>           <dbl>              <dbl>            <dbl>
 1 12        478        456     95.4            5.68               2.13             8796.
 2 1         969        891     92              5.52               2.13            15237
 3 14        272        236     86.8            5.45               2.14            13759
 4 8         616        343     55.7            5.58               1.21             2376
 5 16        214         72     33.6            5.39               1.79            11532.
 6 10        557        152     27.3            5.49               1.93             6103
 7 17        124         13     10.5            5.32               1.66             7006
 8 18        112          3      2.7            5.62               0.822            3174
 9 19         91          2      2.2            5.27               0.953            4998
10 4         757          5      0.7            5.48               1.29            12384
11 0        1262          8      0.6            5.35               1.04             3468.
12 13        307          1      0.3            5.64               0.944            3232
13 5         725          1      0.1            5.33               1.43             2831
14 11        482          0      0             NA                 NA                NA
15 15        255          0      0             NA                 NA                NA
16 2         902          0      0             NA                 NA                NA
17 20         63          0      0             NA                 NA                NA
18 3         795          0      0             NA                 NA                NA
19 6         677          0      0             NA                 NA                NA
20 7         649          0      0             NA                 NA                NA
21 9         571          0      0             NA                 NA                NA
```

Hide

```
# Mapping high spots in clusters 1, 14 and the rest of the section.
mbp_colours <- c(
  "Cluster 1" = "#D55E00",
  "Cluster 14" = "#0072B2",
  "Other MBP-high" = "#756BB1",
  "Below cutoff" = "#D9D9D9"
)

mbp_spots$mbp_region <- factor(
  case_when(
    !mbp_spots$mbp_high ~ "Below cutoff",
    mbp_spots$cluster %in% focus ~ paste("Cluster", mbp_spots$cluster),
    TRUE ~ "Other MBP-high"
  ),
  levels = names(mbp_colours)
)

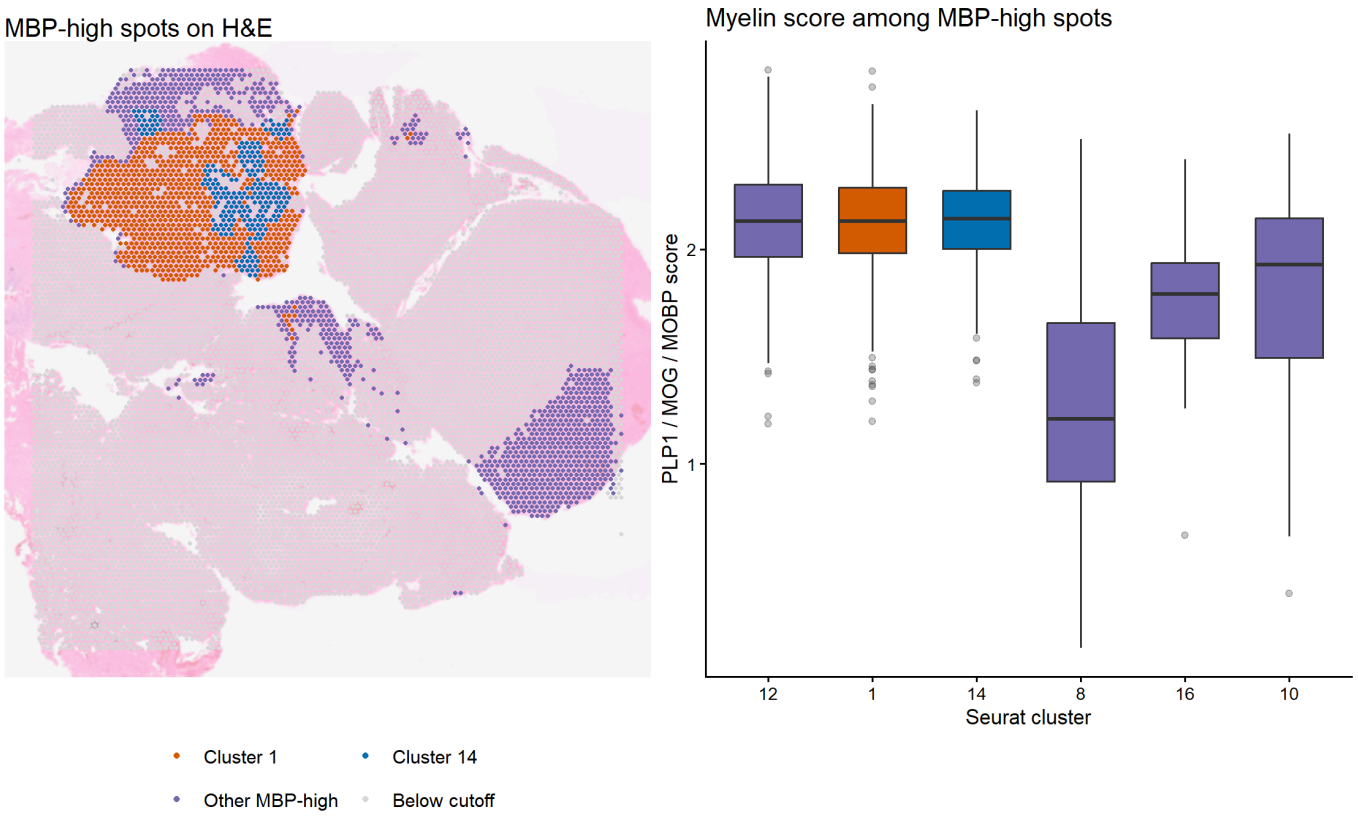
gbm_mbp_map <- AddMetaData(gbm_myelin, metadata = mbp_spots["mbp_region"])

p_mbp_map <- SpatialDimPlot(
  gbm_mbp_map,
  group.by = "mbp_region",
  images = Images(gbm_mbp_map)[1],
  image.scale = "lowres",
  cols = mbp_colours,
  pt.size.factor = 2.5,
  image.alpha = 0.55,
  combine = FALSE
)[[1]] +
  labs(title = "MBP-high spots on H&E", fill = NULL) +
  guides(fill = guide_legend(nrow = 2, byrow = TRUE)) +
  theme(
    legend.position = "bottom",
    legend.text = element_text(size = 10),
    legend.margin = margin(0, 0, 0, 0)
  )

# Comparing the score in clusters with enough high spots to show a distribution.
plot_clusters <- mbp_summary %>%
  filter(high_spots >= 20 | cluster %in% focus) %>%
  pull(cluster)

p_mbp_compare <- mbp_spots %>%
  filter(mbp_high, cluster %in% plot_clusters) %>%
  mutate(cluster = factor(cluster, levels = plot_clusters)) %>%
  ggplot(aes(cluster, MyelinCheck1, fill = mbp_region)) +
  geom_boxplot(width = 0.65, outlier.alpha = 0.25) +
  scale_fill_manual(values = mbp_colours, guide = "none") +
  labs(
    title = "Myelin score among MBP-high spots",
    x = "Seurat cluster",
    y = "PLP1 / MOG / MOBP score"
  ) +
  theme_classic()
```

```
p_mbp_map + p_mbp_compare
```




From the looks of it, cluster 12 demonstrates particularly similar Myelin scores among MBP-high spots and interestingly, at a fairly further region than cluster 1 and 14.

Posteriorly I will investigate what is the major difference between these two areas that drive the correlation score between them to be

0.45725566 (g1 vs g12) 0.43404108 (g12 vs g14)

Particularly when compared to the similarity if g1 vs g14

0.94376386 (g1 vs g14)

Hide

```
library(dplyr)

clusters <- c("1", "14", "12")
myelin_genes <- c("MBP", "PLP1", "MOG", "MOBP")

sct_residuals <- SeuratObject::LayerData(
  gbm_myelin, assay = "SCT", layer = "scale.data"
)

cluster_by_spot <- setNames(
  as.character(gbm_myelin$seurat_clusters),
  colnames(gbm_myelin)
)

spots_by_cluster <- split(
  colnames(sct_residuals),
  factor(cluster_by_spot[colnames(sct_residuals)], levels = clusters)
)

mbp_high_spots <- rownames(mbp_spots)[which(mbp_spots$mbp_high)]

data.frame(
  cluster = clusters,
  all_spots = lengths(spots_by_cluster),
  MBP_high_spots = lengths(lapply(spots_by_cluster, intersect, y = mbp_high_spots))
)
```

| | cluster<br><chr> | all_spots<br><int> | MBP_high_spots<br><int> |
|---|---|---|---|
| 1 | 1 | 969 | 891 |
| 14 | 14 | 272 | 236 |
| 12 | 12 | 478 | 456 |

3 rows

Hide

```
mean_profiles <- function(barcodes) {
  profiles <- vapply(
    spots_by_cluster,
    function(spots) {
      selected <- intersect(spots, barcodes)
      Matrix::rowMeans(sct_residuals[, selected, drop = FALSE])
    },
    numeric(nrow(sct_residuals))
  )
  rownames(profiles) <- rownames(sct_residuals)
  profiles
}

all_profiles <- mean_profiles(colnames(sct_residuals))
high_profiles <- mean_profiles(mbp_high_spots)
held_out <- !rownames(all_profiles) %in% myelin_genes

pairwise_rho <- function(profiles) {
  rho <- cor(profiles, method = "spearman", use = "pairwise.complete.obs")
  c(
    "12 vs 1" = rho["12", "1"],
    "12 vs 14" = rho["12", "14"],
    "1 vs 14" = rho["1", "14"]
  )
}

round(rbind(
  "All spots" = pairwise_rho(all_profiles),
  "All spots excluding myelin markers" = pairwise_rho(all_profiles[held_out, ]),
  "MBP-high spots" = pairwise_rho(high_profiles),
  "MBP-high spots excluding myelin markers" = pairwise_rho(high_profiles[held_out, ])
), 3)
```

```
                                    12 vs 1 12 vs 14 1 vs 14
All spots                             0.457    0.434   0.944
All spots excluding myelin markers    0.455    0.432   0.944
MBP-high spots                        0.493    0.468   0.939
MBP-high spots excluding myelin markers   0.491    0.465   0.939
```

Hide

```
# Find shared positive residuals and genes ranked differently in cluster 12.
gene_profiles <- tibble::tibble(
  gene = rownames(all_profiles),
  C1 = all_profiles[, "1"],
  C14 = all_profiles[, "14"],
  C12 = all_profiles[, "12"],
  shared_residual = pmin(C1, C14, C12),
  rank_shift_vs_1 = rank(C12) - rank(C1),
  rank_shift_vs_14 = rank(C12) - rank(C14)
)

shared_genes <- gene_profiles %>%
  filter(shared_residual > 0) %>%
  slice_max(shared_residual, n = 8, with_ties = FALSE) %>%
  select(gene, C1, C14, C12)

shifted_12 <- gene_profiles %>%
  filter(rank_shift_vs_1 * rank_shift_vs_14 > 0) %>%
  mutate(rank_shift = pmin(abs(rank_shift_vs_1), abs(rank_shift_vs_14))) %>%
  slice_max(rank_shift, n = 10, with_ties = FALSE) %>%
  select(gene, C1, C14, C12, rank_shift_vs_1, rank_shift_vs_14)

shared_genes
```

| gene<br><chr> | C1<br><dbl> | C14<br><dbl> | C12<br><dbl> |
|---|---|---|---|
| PLP1 | 9.820366 | 9.302750 | 9.036538 |
| MBP | 6.455420 | 5.986841 | 8.418800 |
| MOBP | 5.149794 | 5.004131 | 6.675297 |
| CARNS1 | 4.121137 | 3.737067 | 3.880890 |
| CNP | 3.995665 | 3.699561 | 4.792416 |
| TMEM144 | 3.704167 | 3.476139 | 4.027070 |
| CLDND1 | 3.803625 | 3.888989 | 3.473692 |
| ERMN | 3.790241 | 3.470763 | 4.530164 |

8 rows

Hide

```
shifted_12
```

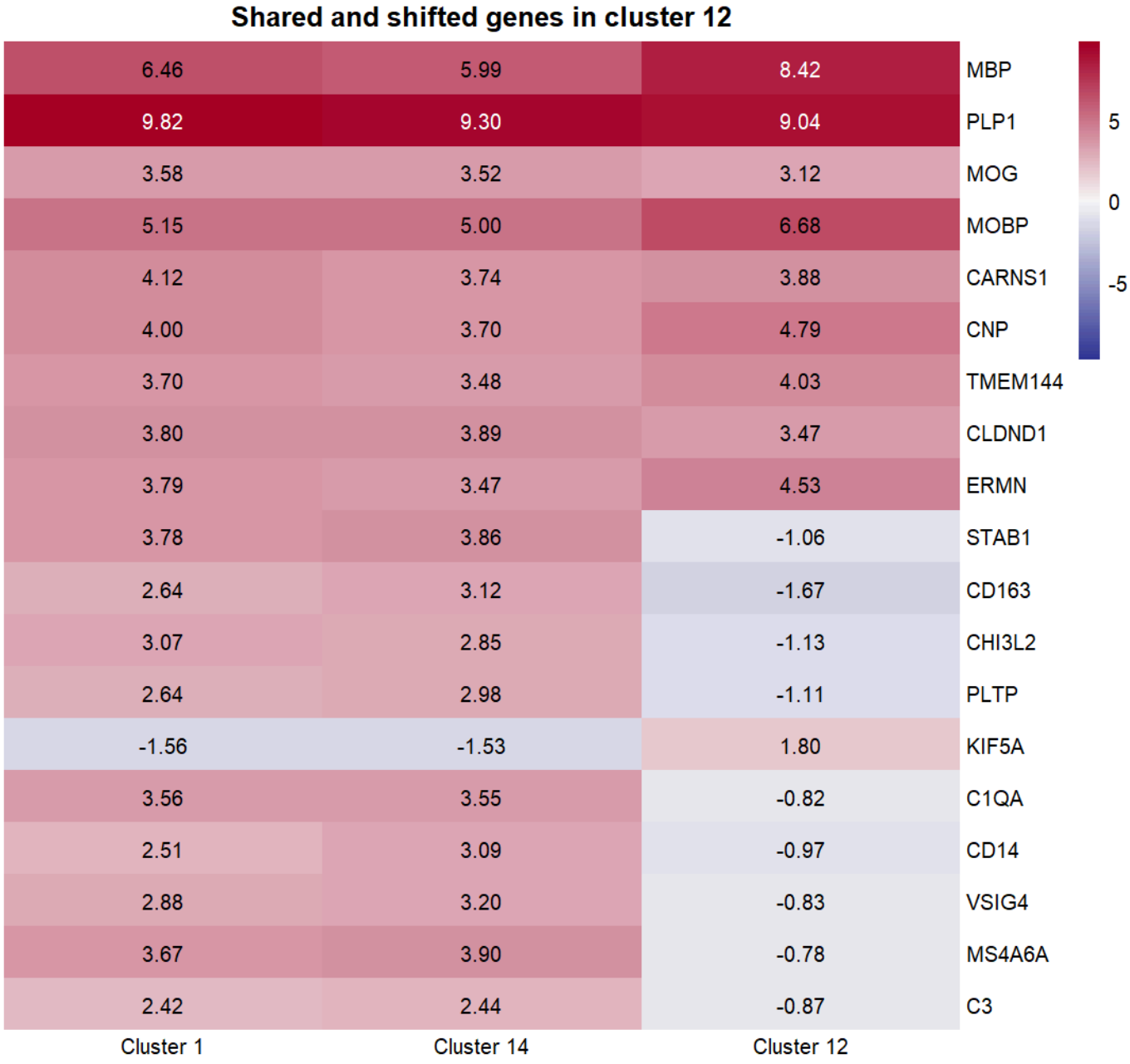
| gene<br><chr> | C1<br><dbl> | C14<br><dbl> | C12<br><dbl> | rank_shift_vs_1<br><dbl> | rank_shift_vs_14<br><dbl> |
|---|---|---|---|---|---|
| STAB1 | 3.775834 | 3.855923 | -1.0602476 | -2950 | -2949 |
| CD163 | 2.643443 | 3.121113 | -1.6678239 | -2948 | -2961 |
| CHI3L2 | 3.070306 | 2.854283 | -1.1255423 | -2946 | -2930 |

| gene<br><chr> | C1<br><dbl> | C14<br><dbl> | C12<br><dbl> | rank_shift_vs_1<br><dbl> | rank_shift_vs_14<br><dbl> |
|---|---|---|---|---|---|
| PLTP | 2.640767 | 2.975190 | -1.1105233 | -2922 | -2931 |
| KIF5A | -1.556623 | -1.533512 | 1.8019306 | 2914 | 2915 |
| C1QA | 3.557624 | 3.545940 | -0.8151792 | -2904 | -2903 |
| CD14 | 2.513616 | 3.094833 | -0.9716438 | -2901 | -2920 |
| VSIG4 | 2.884711 | 3.196468 | -0.8308524 | -2891 | -2895 |
| MS4A6A | 3.669975 | 3.897608 | -0.7766367 | -2890 | -2893 |
| C3 | 2.424955 | 2.438309 | -0.8699562 | -2878 | -2875 |

1-10 of 10 rows

Hide

```
p_cluster_12 <- pheatmap::pheatmap(
  heatmap_values,
  cluster_rows = FALSE,
  cluster_cols = FALSE,
  labels_col = paste("Cluster", clusters),
  angle_col = "0",
  color = colorRampPalette(c("#313695", "#F7F7F7", "#A50026"))(101),
  breaks = seq(-colour_limit, colour_limit, length.out = 102),
  border_color = NA,
  display_numbers = TRUE,
  number_format = "%.2f",
  fontsize_number = 10,
  number_color = number_colours,
  main = "Shared and shifted genes in cluster 12"
)
```




Hide

```
ggplot2::ggsave(
  filename = "GBM_Cluster_12_Comparison_Numbered.png",
  plot = p_cluster_12$gtable,
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# Conclusions between cluster 1, 12 and 14

Cluster 1 and 14 present very similar genetic composition particularly within their predominant set of Myelin score.

Even though cluster 12 presents a fairly similar Myelin score within the MBP predominant spots of this tissue, this cluster demonstrates a shift towards lower genetic expression of genes that present higher SCT transformed RNA expression within cluster 1 and 14.

Interestingly a majority of cells within cluster 12 are actively expression KIF5A a marker that has previously been associated to correlate negatively with tumour grade, with lower levels of expression being highly present in GBM tumour cells (Joyce & Lindsay, 2024).

This could be a good explanation on why these clusters present such high similarity but also, somewhat dissimilar, with a high chance of a GBM tumour slowly forming within cluster 1 and 14, while cluster 12 cells demonstrate to be closely within healthy states.

Another major factor that may indicate this is the highly expression of CD14 and CD163 cells within cluster 1 and 14, indicating direct activity of immune system in this region and possibly part of a forming GBM.

# Comparison of clusters 1, 12 and 14

The comparison began with spots classified as MBP-high using the section-wide 80th percentile. Within this subset, median module scores for PLP1, MOG and MOBP were 2.13 in cluster 1, 2.14 in cluster 14 and 2.13 in cluster 12. The three clusters therefore shared a similar myelin-associated signal.

To compare their broader expression profiles, genes were ranked by their mean SCT residual within each cluster. Spearman correlations were calculated using all spots and then recalculated within the MBP-high subset.

The relative similarity of the clusters persisted after removing MBP, PLP1, MOG and MOBP from the correlation. The agreement between clusters 1 and 14 therefore extended beyond the selected myelin markers.

The gene-level comparison showed lower mean SCT residuals for CD14, CD163, C1QA and VSIG4 in cluster 12. For example, the CD163 residual was −1.67 in cluster 12, compared with +2.64 in cluster 1 and +3.12 in cluster 14. KIF5A showed the reverse pattern, with a mean residual of +1.80 in cluster 12 and negative values in both other clusters.

# Final Conclusions

## Collagen enrichment in clusters 5 and 9 and hypoxia in cluster 7

Clusters 5 and 9 shared elevated expression of COL1A1, COL1A2 and COL3A1. Their cluster-average SCT profiles showed moderate agreement in gene ranks (Spearman ρ = 0.571). Within that shared collagen programme, COL6A3 was detected more frequently in cluster 5 than in cluster 9 (88.7% versus 45.7% of spots).

The hypoxia-associated module score overlapped cluster 7. The marker analysis supported this spatial pattern. CA9 was detected in 80.3% of cluster 7 spots and 7.9% of spots outside the cluster. The corresponding figures for HILPDA were 98.0% and 28.3%.

The collagen-associated expression is consistent with extracellular matrix remodelling. Published spatial studies have described collagen-rich perivascular regions in recurrent GBM and linked hypoxia-associated expression to tissue organisation around necrotic cores (Onubogu et al., 2024; Greenwald et al., 2024). Comparing the score maps with the matched H&E would help establish how the collagen-enriched regions and the cluster 7 hypoxia signal are positioned relative to vessels and necrotic borders in this section.

# Myelin-rich regions in clusters 1 and 14 compared with cluster 12

The higher residuals for myeloid and complement-associated genes in clusters 1 and 14 suggest a stronger myeloid-associated RNA signal in those regions (Miller et al., 2025). The higher KIF5A residual in cluster 12 is compatible with a greater neuronal RNA contribution given the gene's role in axonal transport. Joyce and Lindsay (2024) reported an inverse association between KIF5A expression and glioma grade in a separate cohort analysis.

These spot-level differences may reflect variation in cell composition, transcriptional state or both within this GBM section. Matched histology and tumour-associated copy-number profiles would help assess whether the myelin-rich regions differ in local tumour infiltration.

# References

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